

VIMAAN Run Sheet

One slide per person, in build order, so everyone speaks to what they own. Times are hard — the clock is the enemy, not the judges. Rehearse three times against a timer before you sleep.

The seven minutes

390s SPEAKING + 5s × 6 HANDOFFS

01	02	03	04	05	06
0:00		1:45		3:30	
5:15		7:00			
CLOCK	SEC	SPEAKER	SLIDE & THE ONE THING TO LAND		
01	0:00 → 0:35	35s	Harshit	Title & the frame	
		PITCH LEAD · CAPTAIN	"VIMAAN tells the RBI whether to raise interest rates."		
02	0:40 → 2:05	85s	Soumya	An airfare is a surface	
		DATA ENGINEER	80 cells — one route, on one day.		
03	2:10 → 3:30	80s	Sankalp	Technical approach & index engine	
		INDEX STATISTICIAN	+12 index points of pure drift.		
04	3:35 → 4:40	65s	Bhumika	Feasibility, risk & compliance	
		ACQUISITION LEAD	Two of three lanes need no scraping.		
05	4:45 → 5:55	70s	Shruti	Impact & benefits	
		FRONTEND	9.43% → repo rate → 140 crore people.		
06	6:00 → 6:55	55s	Daksh	Evidence base & close	
		BACKEND · SDMX	"Their published estimators, not ours."		
7:00	420s	TOTAL	390s speaking · 30s handoffs · 5s spare		

Viva protocol

3 MINUTES = 4-6 QUESTIONS

01 One captain routes Harshit takes every question first and hands off by name: "That's Sankalp, our index statistician." Never two people answering at once — it reads as a team that doesn't know who owns what.	02 25 seconds, then stop Number first, then the reason, then silence. If they want more they'll ask. Talking past the answer is how you hand them the next objection.
03 Never invent a number If you don't have it: "We haven't measured that yet — the design is X, and we'd validate against Y." A judge forgives a gap. Nobody forgives a fabricated statistic.	04 Reframe, don't defend Agree with the true part, then redirect. "You're right that dynamic pricing is brutal — that's exactly why a single monthly check is meaningless, and why we use multilateral methods."
05 Say the institution's name ONS, MoSPI, DGCA, the CPI Manual, Rule 135(2). Every named source converts an opinion into a citation. Government panels score citations.	06 Never say "we didn't have time" Say what exists and what's designed: "Lanes A and B are running; Lane C is built and rate-budgeted." Scope honestly, frame it as roadmap, not shortfall.

SMART INDIA HACKATHON 2026

SMART INDIA
HACKATHON
2026

- Problem Statement ID – SIH26056
- Problem Statement Title – Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI)
- Theme – Travel & Tourism
- PS Category – Software
- Team Name – Bharat Bytes

VIMAAN Validated Index of Market Airfares
for Augmenting National-statistics
Skyscanner tells you what a ticket costs. VIMAAN tells the RBI whether to raise interest rates.



500+
ROUTES, 3x A DAY

78 → 500+
VS DGCA TODAY

SDMX
FEED INTO CPI 2024



OPENING LINE – SAY THIS EXACTLY

"Skyscanner tells you what a ticket costs. **VIMAAN** tells the Reserve Bank whether to raise interest rates."

1 **5S**

Problem statement **SIH26056**, for the Ministry of Statistics. We are **Bharat Bytes**.

2 **10S**

"We are not building a fare-search product. We are building **statistical infrastructure** — a daily, audit-grade airfare price index that plugs into the CPI 2024 series."

3 **10S · POINT BOTTOM-LEFT**

The three numbers: **500+ routes**, **three collections a day**, delivered as a machine-readable **SDMX feed** into CPI 2024.

LAND The one-liner. If they remember nothing else, they remember "**tells the RBI whether to raise rates.**" Pause a full beat after it.

HANDOFF "Here's why today's number needs fixing." → Soumya

AVOID Don't read the problem statement title aloud — it's on screen and it burns 15 seconds you don't have.



OPENING LINE — SAY THIS EXACTLY

"Today, MoSPI collects the CPI airfare by hand. DGCA checks 78 routes, once a month, by opening airline websites. That is one fare, per route, per month — about sixty days late."

1 15S · LEFT IMAGE

Manual monitoring, one check, one route. The red bar at the bottom is your line: **incomplete data → biased CPI airfare component → poor policy decisions.**

2 20S · PIVOT RIGHT — THE SLIDE'S JOB

"But an airfare is not a price, it is a **surface**." Same route, Mumbai-Delhi. Price moves with departure date, with how many days ahead you book, with cabin, carrier, stops, baggage.

3 10S · POINT AT THE GRID

"This grid is **eighty cells** — **one route, on one day**. A monthly manual check samples **one cell** of it."

4 20S

"VIMAAN reads the whole surface: **eight lead-time buckets, L1 to L60**, every departure date in a 60-day horizon, three times a day, across 500+ routes."

5 15S · BOTTOM STRIP, FAST

A surface not a price · Jevons and GEKS-Jevons, not a raw average · compliance as architecture · every number reproducible.

LAND 80 cells — **one route, on one day**. This single sentence is why the project exists. Say it slowly.

HANDOFF "So how do we actually build that?" → Sankalp

AVOID Don't read the four innovation cards one at a time — name them in one breath and move. This is your longest slot; the surface explanation is what deserves the seconds.



OPENING LINE — SAY THIS EXACTLY

"Five stages: **collect, clean, analyse, index, publish**. Fully automated — no human in the loop."

1 12S · LEFT, COLUMNS 1-2

Three lanes — statutory tariff pages, licensed airline APIs, public portals — plus DGCA traffic, ATF fuel price, festival calendar. Collections at **02:00, 11:00 and 19:00 IST**.

2 18S · COLUMNS 3-4

Duplicates, one currency, all-inclusive fares, outliers. Then a **hedonic model** strips out stops, duration, departure slot, cabin, baggage — "so when an airline swaps a nonstop for a one-stop, that shows up as a **quality change, not deflation**."

3 30S · RIGHT PANEL — SLOW DOWN, YOUR MOMENT

"Same fares, two methods. Chain a daily average over dynamic pricing and you manufacture **twelve index points of drift** — inflation that never happened. **GEKS-Jevons** on a 25-month rolling window removes it."

4 10S

Jevons geometric mean at the elementary level; **DGCA passenger weights** above it.

5 10S · COLUMN 6

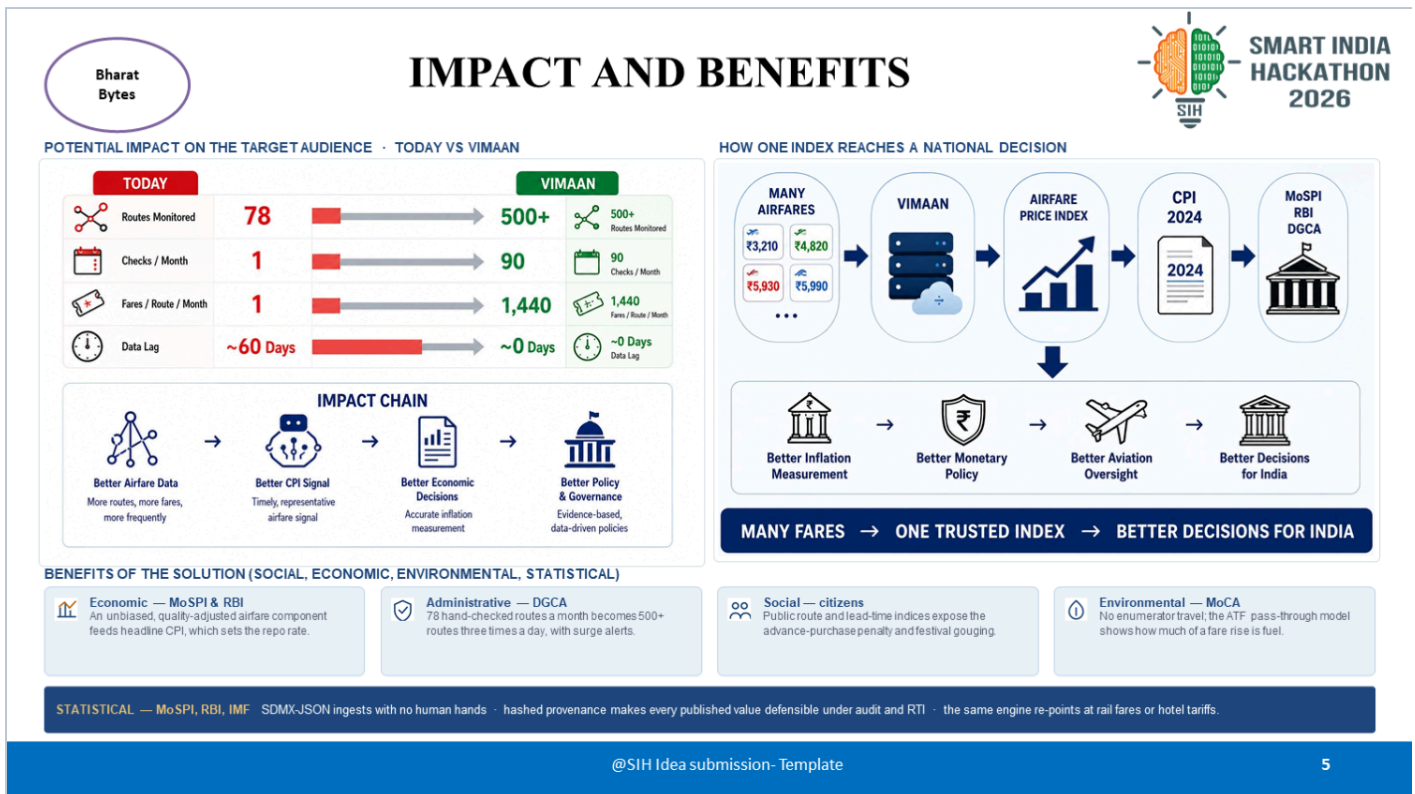
Publish: FastAPI, SDMX-JSON to MoSPI, live dashboard, fare-surge alerts to DGCA.

LAND +12 index points of pure drift. This is the technical claim that separates you from every other team in the room.

HANDOFF "So — can we actually build this in the real world?" → Bhumika

AVOID Do not read the formulae aloud. They're on the slide for the judge who wants them; reading them costs 20 seconds and loses everyone else.





OPENING LINE — SAY THIS EXACTLY

"Four numbers. Routes: **78 to 500-plus**. Checks a month: **1 to 90**. Fares per route per month: **1 to 1,440**. Lag: **sixty days to zero**."

1 15S · RIGHT IMAGE

The chain: many airfares → VIMAAN → one airfare price index → CPI 2024 → MoSPI, RBI, DGCA.

2 20S · THE STRONGEST 20S IN THE DECK

"Air transport sits inside the Transport group, **9.43% of CPI 2024**. Headline CPI sets the repo rate. The repo rate sets EMIs and deposit rates for **140 crore people**. A thin, lagged airfare sample propagates all the way through." **Then pause.**

3 10S

DGCA gets 500+ routes three times daily with automatic surge alerts, instead of 78 by hand.

4 12S

Citizens get a public **lead-time index** — the true advance-purchase penalty. "India publishes no such statistic today."

5 10S

"And it isn't only airfares — swap the collectors and the weights, the same engine gives you a rail-fare or hotel-tariff index."

LAND 9.43% → repo rate → 140 crore people. Say it as one causal sentence, then stop for a beat and let it sit.

HANDOFF "Every claim we've made is sourced —" → Daksh

AVOID Don't claim VIMAAN "fixes inflation". You improve one input to one component. Overclaiming is exactly what gets torn apart in viva.



RESEARCH AND REFERENCES

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

EVIDENCE BASE FOR VIMAAN

A · INDIAN STATUTORY & STATISTICAL BASIS

- MoSPI — FAQs on the CPI 2024 Series, Annexure V
airfare prices are collected through online platforms mospi.gov.in
- PIB — first CPI release on base 2024 = 100, 12 Feb 2026
358 weighted items · COICOP 2018 · Transport weight 9.43% PRID 2227012
- MoSPI — Expert Group Report on updation of the CPI
weight derivation and item-basket methodology mospi.gov.in
- Aircraft Rules, 1937 — Rule 135(2)
every airline shall publish its established tariff on its website civilaviation.gov.in
- Collection of Statistics Act, 2008
statutory mandate for official price collection indiaincode.nic.in
- PIB / MoCA — DGCA Tariff Monitoring Unit
78 routes, monthly, read by hand from airline websites PRID 1810467
- DGCA — Monthly Domestic Traffic Statistics
route-wise passengers and load factors — our index weights dgca.gov.in

B · INTERNATIONAL METHOD & INDEX THEORY

- ONS — aggregate index of air fares (FOI-2023-1164)
fixed advance-purchase windows; fares recorded in month of departure ons.gov.uk
- ONS — Methodology Working Paper 12
index-number methods compared on web-scraped price data ons.gov.uk
- ONS — multilateral index methods in consumer price statistics
GEKS-Jevons, Time-Product-Dummy, rolling windows, splicing ons.gov.uk
- ILO / IMF / OECD / UNECE / Eurostat — CPI Manual 2020
elementary aggregate formulae, why Jevons, not Carli or Dutot imf.org
- Statistics Canada — Air Transportation Index in the CPI, 2020
treating dynamic airline pricing inside an official index statcan.gc.ca
- Statistics Netherlands / Eurostat — web scraping in price statistics
25-month window guidance and chain-drift evidence unece.org
- IOCL — Aviation Turbine Fuel price circulars
input to the fuel pass-through model iocl.com

C · DATA, VALIDATION & WHAT MAKES IT DIFFERENT

Datasets assembled before hour zero
routes_top500.csv(DGCA) · dgca_pax_weights.csv · base_2024_fares.csv
(backcast) · airports_by_region (OpenFlights) · air_prices.csv (IOCL) ·
festival_calendar.csv · backup_fares_21d.parquet

How we prove the index is right
Benchmark against MoSPI's own manually collected airfare series over overlapping
months · cross-lane divergence tests (statutory vs API vs portal) with a flag
threshold · leave-one-route-out sensitivity on the DGCA weights · a published
revision policy.

WHAT SEPARATES THIS FROM A FARE-SEARCH APP

- ✓ A price surface, not a price
- ✓ Jevons + GEKS—Jevons, not a raw average
- ✓ Departure-month attribution, DGCA-weighted
- ✓ An SDMX feed, not just a dashboard
- ✓ A reproducibility hash on every value
- ✓ Two of three lanes are statutory, not scraped
- ✓ Benchmarked against MoSPI's own airfare series

@SIH Idea submission- Template6

OPENING LINE — SAY THIS EXACTLY

"Everything we've claimed is sourced. Three branches — Indian statutory basis, international method, and our own validation."

- 110S · BRANCH A — DO NOT READ ALL 14
- MoSPI's CPI-2024 FAQ, the PIB release on base 2024, Rule 135(2), the Collection of Statistics Act, DGCA traffic data.
- 215S · BRANCH B — THE CREDIBILITY BEAT
- ONS air-fares methodology, ONS on multilateral index methods, the ILO/IMF CPI Manual, Statistics Canada, Statistics Netherlands. "These are the national statistics offices that already run web-scraped price indices — we're using their published estimators, not inventing our own."
- 310S · BRANCH C
- Benchmarked against MoSPI's own manual airfare series, cross-lane divergence tests, a published revision policy.
- 418S · THE CLOSE — FOUR BEATS, SLOW
- "A price surface, not a price. GEKS-Jevons, not an average. An SDMX feed, not just a dashboard. A reproducibility hash on every published number."

LAND	"We're using their published estimators, not inventing our own." This is the credibility line for a government panel.
HANDOFF	"We're Bharat Bytes. Thank you." — then stop talking and stand still.
AVOID	No "that's all from our side", no nervous trailing off. Four beats, team name, thank you, silence.

Tier 1 — Basics

ANYONE ON THE PANEL CAN ASK THESE

1 What is VIMAAN, in one sentence?

Harshit

A real-time airfare price index for India. We collect fares automatically from airline and travel portals across 500+ routes three times a day, and reduce them to **one audit-grade index number** MoSPI can feed into the CPI 2024 airfare component.

2 What problem are you actually solving?

Soumya

MoSPI still collects the CPI airfare by hand from online platforms, and DGCA's Tariff Monitoring Unit checks only **78 routes, once a month**. That's one price from a surface with thousands of cells — thin, lagged about 60 days, and biased.

3 How is this different from MakeMyTrip or Skyscanner?

Harshit

They sell tickets; we produce an official statistic. Their output goes into a shopping cart, ours goes into the CPI. Concretely: a **methodology endpoint**, an **SDMX-JSON feed** and a **reproducibility hash** — none of which a fare-search product has any reason to build.

4 What is a price index, in one line?

Sankalp

A number measuring how a fixed basket's price has changed against a base period set to 100. So **118.7 means fares are 18.7% above the 2024 base** — not that a ticket costs ₹118.70.

5 Why airfares specifically? Isn't it a small item?

Shruti

Because it's the **worst-measured item in the basket**. Air transport sits inside Transport, weighted 9.43% in CPI 2024, and it's where the sampling method is furthest from market reality — one monthly manual check against fares that move 15–20 times a day.

6 Who actually uses this?

Shruti

Four users. **MoSPI** takes the index into CPI. **RBI** gets a cleaner inflation input. **DGCA** gets 500+ routes monitored with surge alerts instead of 78 by hand. **Citizens** get a public lead-time index showing the real advance-purchase penalty.

7 What does "augmenting the CPI" mean?

Daksh

We don't replace the CPI. We replace one **input** to it — the manually collected airfare price — with a daily, quality-adjusted, weighted index, delivered in SDMX-JSON, the format MoSPI's systems already exchange statistics in.

8 So your deliverable is a dashboard?

Daksh

The dashboard is the shop window. The deliverable is the **API**: `/v1/index/national`, `/v1/sdmx/CPI_AIRFARE`, `/v1/methodology/{id}`. Every response embeds coverage, imputation share, revision flag and reproducibility hash. Nothing is published without provenance.

9 How often does it update?

Bhumika

Collection three times daily — **02:00, 11:00, 19:00 IST** — to catch intra-day revenue-management moves. Daily route price is the **median** of intra-day observations. Index publishes daily; the official bulletin is monthly.

10 Where does the number 78 come from?

Bhumika

DGCA's Tariff Monitoring Unit — it monitors airfares on **78 routes, monthly**, by staff opening airline websites. **PIB / MoCA, PRID 1810467**. That's our published baseline to beat.

Tier 2 — Data, legality & engineering

THE OBVIOUS FOLLOW-UPS

11 Isn't web scraping illegal?

Bhumika

Two of our three lanes don't scrape at all. Rule 135(2) of the Aircraft Rules, 1937 *requires* airlines to publish their established tariff — Lane A reads mandated public disclosure. Lane B uses licensed distribution APIs under contract. Lane C reads publicly displayed prices, honours robots.txt, rate-budgets, zero personal data. We explicitly **refuse** CAPTCHA-solving and fingerprint spoofing.

12	What if a portal blocks your collector? The index survives on any one lane . Lanes A and B are statutory and contractual — they can't be blocked. We hold a rolling buffer, so even total collection failure for days doesn't move the monthly number; it moves the coverage percentage , which we publish.	Bhumika
13	Why not just ask the airlines for their data? Rule 135 already makes them publish it. Voluntary sharing would be aggregated, delayed and self-reported — three things a statistician can't audit. We capture what the consumer actually faces , at the moment they face it.	Bhumika
14	Do you store personal data? What about DPDP? Zero personal data. We collect fares, not passengers — no names, no PNRs, no user-linked searches, no real-session cookies. There is no personal data in the pipeline, so DPDP obligations around personal data don't attach. Documented in COMPLIANCE.md.	Bhumika
15	Where do your index weights come from? DGCA Monthly Domestic Traffic Statistics — route-wise passengers in the base period. A route's weight is its share of base-period passengers, so Delhi–Mumbai counts far more than a thin regional route. That sits inside the CPI 2024 Transport structure.	Sankalp
16	How much data per day? Can you handle it? 500 routes × 60 dates × 8 buckets is over-sampled, so we run a PPS rotating panel driven by DGCA traffic — roughly 150,000 observations a day . Single machine on TimescaleDB. Storage is the cheap part; the index maths is the hard part.	Soumya
17	Why PostgreSQL + TimescaleDB rather than plain SQL or Mongo? Everything we query is a time series over a route × date × lead-time cube. Timescale hypertables give continuous aggregates and compression, so a 24-month index query runs in milliseconds. We also keep Parquet + DuckDB for analytical replay — that's what makes reproducibility cheap.	Soumya
18	What is your medallion architecture actually storing? Bronze — immutable raw snapshot with SHA-256, collector version, timestamp. Silver — normalised fare observations, IATA-coded, tax-inclusive, currency-parsed. Gold — the price-surface cube the index engine reads. Bronze is what makes any number re-derivable years later.	Soumya
19	What happens when a website changes its layout? Parsers are config-driven, not hardcoded — selectors live in config. A schema-drift detector compares field presence and value distributions against the last known-good run and alarms per lane on an SLO. Meanwhile the other two lanes keep the index alive.	Bhumika
20	Does this cover international routes? Domestic first — that's where the CPI weight and DGCA's mandate sit. Architecture is identical for international; the changes are currency handling, lead-time buckets (ONS uses 6/3/1 month long-haul) and the weight source. Configuration change, not a redesign.	Soumya

- 21 **Why a geometric mean (Jevons) instead of a simple average?** Sankalp
 Arithmetic means are **upward-biased when prices are volatile**, and airfares are the most volatile thing in the basket. The international CPI Manual prescribes Jevons for elementary aggregates precisely for this. An arithmetic mean would import that bias straight into headline CPI.
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- 22 **What is chain drift, and why does it matter here?** Sankalp
 Chain a price index period-by-period over prices that bounce between buckets and the index **fails to return to its start when prices do** — it ratchets upward. On dynamic airline pricing that manufactures several percent of inflation a year that never happened. On flat simulated fares, a naive daily-chained index drifts up **+12 index points**.
-
- 23 **Explain GEKS-Jevons.** Sankalp
 Instead of chaining $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots \rightarrow t$, GEKS computes **every possible bilateral comparison** within a window and takes the geometric mean of the paths from 0 to t through each intermediate period. Because it's symmetric and uses all pairs it is **transitive** — drift cancels by construction. Jevons is the bilateral formula underneath.
-
- 24 **Why a 25-month window, and what is splicing?** Sankalp
 25 months = two full seasonal cycles plus one, so every month is compared against its own season. An index must never revise published history, so when the window rolls we **mean-splice**: take the new window's movement and link it onto the existing series. Published NSO practice — ONS and Statistics Netherlands both do this.
-
- 25 **What is hedonic quality adjustment — concretely?** Sankalp
 We regress log fare on stops, log duration, departure slot, refundability, baggage and carrier, giving the price of each attribute. **Example**: IndiGo replaces a nonstop with a one-stop — average fare falls, but that's a worse product, not deflation. The model re-prices everything to a fixed reference spec, so the drop reads as a **quality change**.
-
- 26 **A ticket bought in August for a December flight — which month's inflation?** Sankalp
 International practice (ONS FOI-2023-1164) attributes the fare to the **month of departure**, not purchase, because the index measures the price of consuming the flight. We attribute on that **use basis** and publish an **acquisition-basis series in parallel** so MoSPI can choose. Almost nobody handles this, and it materially changes the answer.
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- 27 **How do you seasonally adjust when Diwali moves every year?** Sankalp
X-13ARIMA-SEATS with moving-holiday regressors for Diwali, Holi, Eid, Christmas and school-vacation windows. A plain seasonal filter assumes the spike falls in the same calendar month every year — for Indian festivals it doesn't, so it misattributes the spike and leaves a phantom seasonal in the wrong month.
-
- 28 **CPI base is 2024 — but you have no 2024 fares. What's your denominator?** Sankalp
 Honest answer: we **backcast it** from DGCA published route yields, archived fare data and web-archive snapshots, and publish the base as **provisional with a confidence band** that narrows as live data accumulates. We flag it rather than hide it — a published uncertainty is defensible, a silent assumption isn't.
-
- 29 **Listed fare versus what people actually pay — isn't your index biased?** Sankalp
 Yes, and we correct rather than pretend. Portals show the cheapest bucket; realised yield differs. We estimate **k = realised yield ÷ listed median** from DGCA route revenue and passenger data, apply it as a slow-moving level correction, and publish uncertainty bands. For an *index*, a stable level bias largely cancels — what matters is the movement.
-
- 30 **Laspeyres, Lowe or Young — which and why?** Sankalp
 Upper level uses a **Lowe/Young** index with DGCA base-period passenger weights, which is what CPIs actually run in practice. We also publish the classic **Laspeyres** form alongside for direct comparability with MoSPI's existing method — and explain the gap rather than quietly presenting one number.
-
- 31 **What if a route has no fares on a given day — missing cells?** Sankalp
 A **two-way fixed-effects imputation** — route FE + date FE + lead-bucket effect — so a missing cell fills from the route's own level and the day's market-wide movement. We publish the **imputation share** with every number, and anything below a coverage threshold is marked **provisional**.

32 **What is Time-Product-Dummy, and why run both it and GEKS?**

Sankalp

TPD is a regression — $\ln p_{jt} = \mu_t + \lambda_j + \varepsilon$ — where the time dummies μ_t are the index, with product fixed effects absorbing quality. It's a second, independent multilateral estimator. We run both because **agreement between two different estimators is evidence**; divergence is a flag that something in the data is wrong.

33 **How would you prove your index is actually right?**

Daksh

Four tests. **One:** benchmark against MoSPI's own manual airfare series over overlapping months. **Two:** cross-lane divergence tests — statutory vs API vs portal — with a flag threshold, so we never silently average disagreeing sources. **Three:** leave-one-route-out sensitivity on the DGCA weights. **Four:** a published revision policy.

34 **Why three collections a day rather than one, or twenty?**

Sankalp

Revenue-management systems reprice through the day, so one snapshot samples an arbitrary point on that path. Three — morning, midday, evening — captures the shape at acceptable request cost, and we take the **median**, robust to a single anomalous scrape. Twenty adds load without materially reducing variance.

35 "Dynamic pricing makes any airfare index meaningless."

Sankalp

"Dynamic pricing is precisely why a **single monthly manual check** is meaningless — it samples one point on a curve that moves twenty times a day. It's not why an index is meaningless. We sample eight lead-time buckets three times daily, take medians, and use multilateral estimators provably drift-free under exactly this volatility. Volatility is the argument *for* our method."

36 "This is just a scraper with a chart on top."

Harshit

"The scraper is about 15% of the work and the least interesting part. The project is the **index engine**: hedonic quality adjustment, Jevons elementary aggregates, GEKS-Jevons multilateral on a rolling window, festival-aware seasonal adjustment, departure-month attribution — then a methodology endpoint and a reproducibility hash on every number. That's a statistical instrument, not a chart."

37 "Could MoSPI actually deploy this? What would it take?"

Daksh

"Technically, today: `/v1/sdmx/CPI_AIRFARE` emits SDMX-JSON, the exchange format their systems already speak — no integration project. Institutionally it needs what any new official series needs: a methodology review, a parallel-run against the existing manual series, and a published revision policy. We built all three in from the start."

38 "What's actually running versus what's on the slide?"

Harshit

Answer **honestly and specifically** — name what's live, what's built-not-scaled, what's designed. Then close: "and the whole thing comes up with `docker compose up`, seeded with 21 days of real pre-collected fares, so it runs with the network unplugged." Judges reward a team that knows exactly where its own edges are.

39 "Why should the government trust a number produced by six students?"

Daksh

"Because it doesn't have to trust us — it can **check us**. Every published value carries coverage, imputation share, a revision flag and a reproducibility hash, and the methodology endpoint returns the exact formula and parameters used for *that number*. Anyone can re-derive it from hashed raw snapshots. And we aren't inventing estimators — these are ONS and CPI Manual methods."

40 "Airfare is a tiny slice of CPI. Why does this matter?"

Shruti

"Two reasons. It's the item where **measurement error is largest relative to weight** — a small weight measured badly still pollutes the aggregate. Second, the platform generalises: swap collectors and weights and the same engine produces a rail-fare or hotel-tariff index. Airfare is the **hardest first instance**, deliberately chosen."

41 "What if airlines game your collection once they know it feeds the CPI?"

Bhumika

"Real risk, and it's why we don't depend on one source. Lane A reads the **statutory published tariff** — misstating that is a regulatory matter, not a scraping problem. Cross-lane divergence tests surface a source that starts disagreeing with the others. And the listed-to-transacted calibration against DGCA revenue is an independent anchor on reality."

42 "What does it cost to run, and who maintains it afterwards?"

Harshit

"One machine plus object storage — everything open source, nothing licensed, nothing to procure. **Marginal cost per added route is effectively zero**, which is why 78 → 500+ isn't a budget question. Operationally it's designed to sit inside DGCA's or MoSPI's existing IT, with Grafana SLOs per lane so a failure pages someone instead of silently corrupting a statistic."

43 "What's the weakest part of your solution?"

Harshit

Never say "nothing". Say: "**The 2024 base period**." We have no live 2024 fares, so we backcast from DGCA yields and archives and publish the base as provisional with a confidence band. It's the one number we'd want a year of parallel running to firm up — which is exactly why we built a published revision policy instead of pretending the base is exact.

44 "Show me the one thing that proves you understand this problem."

Sankalp

Go straight to slide 3, right panel. "**Identical simulated fares. Zero true inflation. The naive daily-chained index drifts up twelve points; GEKS-Jevons stays flat.**" Then: "Most teams will build the red line and not know it's wrong. That single choice is the difference between a working index and a misleading one."

Numbers cheat sheet — every one of these is lookup-able

78	Routes DGCA's Tariff Monitoring Unit checks, monthly, by hand. <i>PIB/MoCA PRID 1810467</i>
500+	Routes VIMAAN monitors, three times a day
9.43%	Transport group weight in CPI 2024 — air transport sits inside it
~150,000	Observations a day on the PPS rotating panel
8 buckets	Lead times L1, L3, L7, L14, L21, L30, L45, L60 days before departure
60 days	Forward departure horizon · also the current data lag we remove
80 cells	8 lead buckets × 10 departure dates — one route, one day
+12 pts	Pure chain drift a naive daily-chained index manufactures on flat fares
25 months	GEKS rolling window — two seasonal cycles plus one
3× daily	02:00 · 11:00 · 19:00 IST
1 → 1,440	Fares per route per month, today vs VIMAAN
Rule 135(2)	Aircraft Rules 1937 — airlines <i>must</i> publish established tariffs
2 of 3	Intake lanes that involve no scraping at all

Tonight, before you sleep

- Three full run-throughs against a timer. Third one standing, no notes.
- Each person memorises **only their opening line and their handoff line**. Beats can be improvised; the seams cannot.
- Tier-4 drill: someone reads hostile questions at random, answers capped at 25 seconds, no exceptions.
- Confirm docker `compose up` works offline on the demo laptop; backup video on two devices.
- Open the deck in presentation mode on the actual machine you'll present from — check the fonts render.

Every number you say should be one a judge could look up. That's the whole posture — you're not pitching an app, you're proposing a statistic.